4.1 $\vec{x} = (1,2,3)$, $\vec{y} = (-1,0,1)$

a) $\|x\| = \sqrt{1^2+2^2+3^2} = \sqrt{14}$

b) $\|x-y\| = \sqrt{(1-(-1))^2+2^2+(3-1)^2} = \sqrt{4+4+4} = 2\sqrt{3}$

c) $\cos\varphi = \dfrac{\vec{x}^T\vec{y}}{\|x\|\cdot\|y\|} = \dfrac{-1+0+3}{\sqrt{14}\cdot\sqrt{2}} = \dfrac{2}{\sqrt{14}\cdot\sqrt{2}} = \dfrac{\sqrt{7}}{7} \Rightarrow \varphi \approx 1{,}183$ rad

4.2 $\|x+y\| \le \|x\| + \|y\|$

$\sqrt{(x_1+y_1)^2+\dots(x_n+y_n)^2} \le \sqrt{x_1^2+\dots+x_n^2} + \sqrt{y_1^2+\dots+y_n^2}$

je rovno pouze když $\vec{x}=\vec{0}$ a/nebo $\vec{y}=\vec{0}$

4.3 báze ortogonálního doplňku $\operatorname{span}\left\{\underset{\vec{x}_1}{\begin{pmatrix}0\\1\\1\end{pmatrix}}, \underset{\vec{x}_2}{\begin{pmatrix}1\\2\\3\end{pmatrix}}\right\}$

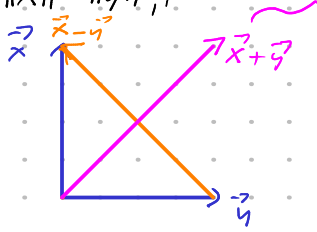
$X = \operatorname{span}\{\vec{x}_1, \vec{x}_2\}$

$\vec{y} \in \mathbb{R}^n$, $y \perp X$

$$y = \vec{x}_1 \times \vec{x}_2 = \begin{pmatrix}0\\1\\1\end{pmatrix} \times \begin{pmatrix}1\\2\\3\end{pmatrix} = \begin{vmatrix} i & j & k \\ 0 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \begin{pmatrix} 1\cdot3-1\cdot2 \\ 1\cdot1-0\cdot3 \\ 0\cdot2-1\cdot1 \end{pmatrix} = \begin{pmatrix}1\\1\\-1\end{pmatrix}$$

4.5

a) $\|x\| = \|y\|$, pak $(x+y) \perp (x-y)$



b) $x \perp y$, pak $\|x\|^2 + \|y\|^2 = \|x-y\|^2$

$$\|x-y\|^2 = (x-y)^T(x-y) = (x^T-y^T)(x-y) = x^Tx - \underset{0}{x^Ty} - \underset{0}{y^Tx} + y^Ty = x^Tx + y^Ty = \|x\|^2 + \|y\|^2 \quad \blacksquare$$

5.13 $x \perp y$, $\operatorname{span}\{x, y\} = \operatorname{span}\left\{\begin{pmatrix}0\\1\\1\end{pmatrix}, \begin{pmatrix}1\\2\\3\end{pmatrix}\right\}$

$$x = \begin{pmatrix}0\\1\\1\end{pmatrix}$$

$$y = \begin{pmatrix}1\\2\\3\end{pmatrix} - v \cdot \vec{x} = \begin{pmatrix}1\\2\\3\end{pmatrix} - v \cdot \begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}1\\2-v\\3-v\end{pmatrix} = \begin{pmatrix}1\\2-\frac{5}{2}\\3-\frac{5}{2}\end{pmatrix} = \begin{pmatrix}1\\-1/2\\1/2\end{pmatrix}$$

$$x^T y = 0 \rightarrow 0 \cdot 1 + 1 \cdot (2 - v) + 1 \cdot (3 - v) = 5 - 2v = 0 \rightarrow v = \frac{5}{2}$$

$$x = \begin{pmatrix}0\\1\\1\end{pmatrix}, y \begin{pmatrix}2\\-1\\1\end{pmatrix}$$